

Data Structures

2. Tuples

The tuples are sequences of Python objects.

To create a tuple, one lists the values, separated by commas:

```
x = 1, 4, 9, 16, 25  
print(x)
```

Output: (1, 4, 9, 16, 25)

It should be noted that tuples are identified by a series of values, surrounded in two brackets.

2.1. Extraction of the Elements

The elements of a tuple are extracted in the same way as those in the lists

```
print(x[0])
```

Output: 1

2.2. Modification

Unlike lists, tuples are inalterable (i.e. cannot be modified after they have been created)

```
x[0] = 1
```

Output:

```
## Error in py_call_impl(callable, dots$args, dots$keywords):  
TypeError: 'tuple' object does not support item assignment  
##  
## Detailed traceback:  
##   File "<string>", line 1, in <module>
```

It is possible to **nest tuples** inside another tuple. To do this, parentheses are used:

```
x = ((1, 4, 9, 16), (1, 8, 26, 64))  
print(x)
```

Output: ((1, 4, 9, 16), (1, 8, 26, 64))